

Patient Shortage in Histopathology: Do Additional Patients Help Without Adding Coverage?

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Abstract

At a fixed patch budget, TCGA-UT patch classifiers trained on more patients are more accurate than redundancy-adjusted effective support predicts. Coverage of tissue source sites explains little of this residual, and a first test of patient coverage stopped before training because its low-coverage allocation covered held-out patients almost as well as a random one. This experiment repeats the test with a low-coverage allocation built around the coverage measure. Patient similarity is the cosine distance between patient-mean Virchow2 embeddings, so all 30 cancer types enter. Three nested allocations share a budget of 160 patches per class: five random seed patients with 32 patches each; the seeds and the fifteen nearest neighbours of a single *focal* seed with eight patches each; and the seeds and fifteen randomly drawn patients with eight patches each. Concentrating all additions around one seed leaves the nearest training patient of most held-out patients unchanged, so the first step quadruples patients while adding little coverage, and the second step adds coverage while patients, patches, and effective support stay fixed. The *coverage gain* compares the two broad allocations, and the *within-concentration residual* measures what the first step adds beyond effective support. A prespecified manipulation check, run before any classifier is trained, requires the concentrated allocation to attain at most half of the random allocation's coverage improvement. A coverage gain above one percentage point with a residual below one point per doubling of patients would identify patient coverage as the missing source of information; the reverse pattern would locate it in properties of additional patients that their mean embedding does not express. The concentrated allocation attains 67 to 73 percent of the random allocation's coverage improvement in the three splits, and only testicular germ cell tumours meet the requirement, so the experiment stops at the manipulation check and both contrasts remain unmeasured. A diagnostic on the same embeddings shows why: for more than half of the held-out patients whose nearest seed is not the focal seed, a neighbour of the focal seed lies closer than their own seed, and these patients supply about twice as much of the leakage as the focal cell. Five further nested rules, including coverage-maximizing farthest-point and facility-location selection, reach leakage no lower than 0.53, because quadrupling patients shortens the nearest-patient distance for most held-out patients whichever patients are added. Coverage still varies strongly between cohorts of equal size, however: twenty patients forming one neighbourhood and twenty patients selected by facility location around the same seed differ in coverage by 84 to 95 percent of the change from five to twenty random patients. A follow-up experiment therefore contrasts these two allocations directly, at a fixed patient count.

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1 Motivation

Spreading a labelling budget over more patients improves TCGA-UT patch classification beyond what redundancy explains. At 160 patches per class, twenty patients contributing eight patches each outperform five patients contributing 32 patches each by 9.52 percentage points of patient-macro balanced accuracy (Queisler, 2026a). *Effective support*, the patch count discounted for within-patient redundancy, accounts for less than half of this advantage: at equal effective support, each doubling of patients still adds 2.62 points (Queisler, 2026b), about 5.2 of the 9.52 points over the two doublings of this comparison. Tissue source sites explain little of this residual: doubling sites at a fixed patient count adds 0.84 points, whereas additional patients from the same sites still add 1.79 points per doubling beyond effective support (Queisler, 2026d). Most of the residual lies in the seventeen smaller cancer types, which the site design could not reach.

Patient coverage, the share of a cancer type’s between-patient variation that the training patients represent, is the remaining candidate that requires no labels (Komura et al., 2022). Its first test compared a broad allocation formed by the nearest neighbours of five seed patients with one formed by randomly drawn patients (Queisler, 2026c). The test stopped at its prespecified manipulation check, before any classifier was trained: the neighbour allocation reached 84 to 88 percent of the random allocation’s coverage improvement. The cause lies in the coverage measure, the mean distance from held-out patients to their *nearest* training patient. Because the seeds are part of both allocations and the neighbours were spread over all five of them, both rules shortened this distance for nearly the same held-out patients. A low-coverage allocation must therefore leave the nearest training patient of most held-out patients unchanged, not merely add patients that resemble the seeds.

Concentrating all additional patients around a single seed meets this requirement. The four other seeds keep their neighbourhoods untouched, so coverage can improve only for held-out patients near the chosen seed. The question stays the one posed by Queisler (2026c): *does a training patient unlike the current cohort contribute more than a patient resembling it, and do additional patients still help beyond effective support when they add little coverage?*

2 Study design

Only the composition of the training allocations changes. The design inherits from Queisler (2026c) the 30 eligible TCGA-UT cancer types, the frozen 2,560-dimensional Virchow2 features (Zimmermann et al., 2024), the three fixed patient-disjoint splits with their natural validation and test partitions, the nested round-robin patch order, and the readout: multinomial logistic regression with its regularization strength selected on the validation partition for every fit. It also inherits the patient distance. For split r and class c , $\bar{\mathbf{x}}_{irc}$ is the mean feature vector of patient i ’s patches of the class, $\boldsymbol{\mu}_r$ the average of these means over all classes, and

$$\mathbf{e}_{irc} = \frac{\bar{\mathbf{x}}_{irc} - \boldsymbol{\mu}_r}{\|\bar{\mathbf{x}}_{irc} - \boldsymbol{\mu}_r\|}, \quad d(i, j) = 1 - \mathbf{e}_{irc}^\top \mathbf{e}_{jrc}. \quad (1)$$

Table 1 lists the allocations. The deep and the broad random allocation are those of Queisler (2026c); only the low-coverage allocation differs.

2.1 Allocation rules

Three rules define the training sets and check the manipulation before any classifier is trained. First, every class is admitted. The pool $\mathcal{E}_{rc}$ contains the training patients with at least 32 patches of class c ; pools hold between 30 and 514 patients, so every class can supply twenty.

Second, the allocations are nested. For each split, class, and draw $d \in \{1, \dots, 5\}$, the five seeds are the patients of the stored five-patient draw of Queisler (2026a), whose predictions are

Allocation	Patients G	Patches per patient m	Patients
Deep	5	32	5 random seeds
Broad, concentrated	20	8	seeds, 15 neighbours of focal seed
Broad, random	20	8	seeds, 15 random patients

Table 1: The three allocations share a budget of 160 patches per class and form two steps. From the deep to the broad concentrated allocation, patient count quadruples while all added patients resemble one seed. From the broad concentrated to the broad random allocation, coverage increases while patients, patches per patient, budget, and effective support stay fixed.

reused for the deep allocation. One seed is drawn uniformly at random as the *focal seed*. The concentrated allocation adds the fifteen patients of $\mathcal{E}_{rc}$ that are not seeds and lie closest to the focal seed under Equation (1), with ties broken by patient identifier. The random allocation adds the same fifteen randomly drawn patients as in Queisler (2026c), so it is distributed like the twenty-patient draw of the breadth–depth grid. With $\mathcal{P}^D$, $\mathcal{P}^C$, and $\mathcal{P}^R$ the patient sets of the three allocations, $\mathcal{P}^D \subset \mathcal{P}^C$, $\mathcal{P}^D \subset \mathcal{P}^R$, and $|\mathcal{P}^C| = |\mathcal{P}^R| = 20$. A seed’s eight patches in either broad allocation are the first eight of its 32 deep patches. The design requires 30 new fits: the broad random fits of Queisler (2026c) were never run, so both broad allocations are fitted for three splits and five draws.

Third, a manipulation check compares how well the allocations cover held-out patients. The *coverage distance* of a patient set is the mean distance from the class’s validation patients $\mathcal{V}_{rc}$ to their nearest training patient,

$$r_{rc}(\mathcal{P}) = \frac{1}{|\mathcal{V}_{rc}|} \sum_{i \in \mathcal{V}_{rc}} \min_{j \in \mathcal{P}} d(i, j). \quad (2)$$

With r_D , r_C , and r_R averaged over classes and draws within a split, the *coverage leakage*

$$\kappa = \frac{r_D - r_C}{r_D - r_R} \quad (3)$$

is the share of the random allocation’s coverage improvement that the concentrated allocation already attains. Fitting proceeds only if $\kappa \leq 0.5$ in all three splits; the class-level κ_c is recorded alongside. The rule targets this quantity directly: a validation patient whose nearest seed is not the focal seed can gain from the concentrated additions only if one of them lies closer to it than that seed, which the concentration around another seed makes unlikely. With one focal seed among five, roughly a fifth of the validation patients can benefit.

3 Evaluation

Both steps of Table 1 are read with the same accuracy measure and the same uncertainty procedure.

3.1 Coverage gain and within-concentration residual

Accuracy is patient-macro balanced accuracy over all 30 classes: patch recall is averaged within each test patient and true class, then across patients within each class, and finally across classes. Averaging three splits and five draws gives A_D , A_C , and A_R in percent. The *coverage gain*

$$\Delta_C = A_R - A_C \quad (4)$$

measures what dissimilar patients contribute at fixed patients, patches per patient, budget, and effective support. Under patient coverage it is positive; under count beyond coverage it is close to zero.

The first step also raises effective support. With $\bar{\rho}_c$ the full-feature intraclass correlation of class c (Queisler, 2026b), effective support is $N_{\text{eff}}(G, m) = \frac{1}{30} \sum_c Gm / (1 + (m - 1)\bar{\rho}_c)$, which gives 10.06 per class for the deep and 36.58 for both broad allocations. The surface $A = \alpha + \beta \log N_{\text{eff}} + \gamma \log G$ is fitted to the stored predictions of the nine-cell breadth–depth grid, and its reference residual $b_{\text{ref}} = \hat{\gamma} \log 2$ must reproduce 2.62 points per doubling. Dividing the unexplained part of the first step by its two doublings of patients gives the *within-concentration residual*

$$b_C = \frac{A_C - A_D - \hat{\beta} \log(N_{\text{eff}}(20, 8)/N_{\text{eff}}(5, 32))}{2} \quad (5)$$

in points per doubling of patients. If coverage carries the residual, b_C falls well below b_{ref} and may turn negative, because effective support counts the concentrated patients as independent of each other. If coverage does not matter, b_C stays close to b_{ref} .

Two secondary analyses describe where a coverage gain arises. First, each test patient is assigned per draw to the *focal cell* if its nearest seed is the focal seed and to the *other cells* otherwise, and Δ_C is computed within both strata with the patient weights of the primary endpoint. Patient coverage predicts a gain concentrated in the other cells, which only the random allocation extends, and no gain or a loss in the focal cell; count beyond coverage predicts similar gains in both. Second, all quantities are reported separately for the thirteen site classes of Queisler (2026d) and the seventeen remaining classes, with the surface refitted to each subgroup, together with the share of concentrated and random additions that come from the focal seed’s tissue source site. These analyses carry no decision threshold.

3.2 Uncertainty

Uncertainty follows Queisler (2026a). Each of 2,000 paired Bayesian bootstrap replicates assigns Dirichlet(1, ..., 1) weights to the distinct test patients (Rubin, 1981), shared across allocations, splits, draws, and classes. The same weights enter the refitted surface, so $\hat{\beta}$, b_{ref} , and b_C remain paired. The 2.5th and 97.5th percentiles give exploratory 95% intervals, including one for $b_C - b_{\text{ref}}$. Correlations and embeddings are held fixed. The standard deviation of each allocation’s accuracy and of both steps across the fifteen split–draw combinations is reported as a descriptive measure of sensitivity to the drawn seeds.

4 Interpretation criteria

Conclusions rest on the test-patient intervals for Δ_C and b_C and on the practical threshold of one percentage point used in the preceding experiments (Table 2).

Patient coverage. If dissimilar patients add a practically relevant gain and concentrated patients add nothing beyond effective support, the missing information on TCGA-UT is coverage of between-patient variation. Coverage distance would then complement effective support as a shortage signal, and a mitigation would select or weight patients for annotation by their distance to the patients already covered.

Count beyond coverage. If dissimilar patients add nothing while the within-concentration residual stays above one point, the benefit of additional patients does not depend on how different their mean appearance is. The missing information would lie in properties a patient mean does not express, such as the distribution of a patient’s patches, and a signal would need a distance between patch distributions.

Account	Claim	Coverage gain Δ_C	Within-concentration residual b_C
Patient coverage	Additional patients help because they differ from the patients already in training	Interval above 1	Interval below 1
Count beyond coverage	Additional patients help even when they resemble the patients already in training	Interval within $[-1, 1]$	Interval above 1
Both	Coverage and patient count each contribute	Interval above 1	Interval above 1

Table 2: Each account predicts a distinct combination of the two contrasts. Intervals are test-patient 95% intervals in percentage points; b_C is expressed per doubling of patients. The coverage account admits a negative b_C . All other combinations, including any interval that crosses a threshold, are inconclusive.

Both and inconclusive outcomes. If both contrasts exceed the threshold, coverage explains part of the residual, and $b_C - b_{\text{ref}}$ quantifies which part; the leakage κ bounds how much coverage the concentrated allocation still adds. All other combinations leave the accounts open.

5 Results

The manipulation check ran for all 30 cancer types, three splits, and five draws per class, and it stopped the experiment before any classifier was trained. Coverage leakage is $\kappa = 0.69, 0.67$, and 0.73 in the three splits, against a permitted maximum of 0.5 . Concentrating all additions around one focal seed lowers leakage from the 0.84 to 0.88 of the neighbour allocation (Queisler, 2026c), but the concentrated allocation still attains about two thirds of the coverage improvement of the random allocation. The coverage gain Δ_C and the within-concentration residual b_C are therefore not estimated, and the accounts of Table 2 remain untested. The check does measure where the coverage improvement arises, and a diagnostic on the same embeddings, reported in Sections 5.3 and 5.4, tests the design’s central premise and several alternative allocation rules before any further fitting.

5.1 Manipulation check

The concentrated allocation shortens the coverage distance of held-out patients by clearly less than the random allocation, but by far more than half as much (Table 3). Averaged over classes and draws, validation patients lie at a mean distance of 0.386 from their nearest patient of the deep allocation. Adding fifteen neighbours of the focal seed shortens this distance to 0.322 , and adding fifteen random patients shortens it to 0.294 . The room between the two broad allocations thus grows from 0.013 in the neighbour design to 0.028 , yet the concentrated allocation still delivers a reduction of 0.062 to 0.068 . The three splits agree closely.

Split	r_D	r_C	r_R	$r_D - r_R$	κ
1	0.387	0.324	0.296	0.091	0.690
2	0.385	0.324	0.293	0.092	0.668
3	0.385	0.317	0.292	0.093	0.734

Table 3: The concentrated allocation attains about two thirds of the coverage improvement of the random allocation, so the manipulation check of Section 2.1 fails in every split. Coverage distances are mean cosine distances from validation patients to their nearest training patient, averaged over the 30 classes and five draws; κ is the coverage leakage of Equation (3), which the design requires to be at most 0.5.

5.2 Leakage by cancer type

Concentration lowers leakage in almost every cancer type, but only one reaches the intended contrast. Averaged over splits, class-level leakage has a median of 0.69, down from 0.86 for the neighbour allocation, and ranges from 0.38 for testicular germ cell tumours to 1.04 for uveal melanoma (Table 4). Across the 90 combinations of classes and splits, six fall to 0.5 or below: testicular germ cell tumours in all three splits, and rectum adenocarcinoma, lung adenocarcinoma, and thyroid carcinoma in one split each. Six combinations exceed one: uveal melanoma and uterine carcinosarcoma in two splits each, and ovarian serous cystadenocarcinoma and rectum adenocarcinoma in one split each. Leakage stays high in small and large classes alike, as it did for the neighbour allocation.

Concentration restricts institutions only mildly. On average, 27 percent of the concentrated additions share the focal seed’s tissue source site, against 17 percent of the random additions. Two classes with few contributing sites, adrenocortical carcinoma and pheochromocytoma and paraganglioma, reach about 70 percent under both rules, so the difference between the rules there is nil. The overlap with site coverage that Section 7 anticipates is thus small compared with the failure of the coverage manipulation.

5.3 Why concentration does not withhold coverage

The design rests on one premise: a validation patient whose nearest seed is not the focal seed should gain nothing from the concentrated additions. A diagnostic on the same embeddings, draws, and allocations tests this premise directly. It reproduces the coverage distances of Table 3 exactly and splits each allocation’s improvement $r_D - r$ into the contributions of validation patients in the focal cell and in the other cells, each expressed as a share of the random allocation’s total improvement $r_D - r_R$ (Table 5).

The focal cell behaves as intended but is small. It holds 17 to 23 percent of the validation patients, close to the one fifth expected from five seeds, and concentrated additions improve coverage there only slightly more than random additions do, contributing 21 to 29 percent of the random allocation’s improvement against 18 to 25 percent. The premise fails in the other cells. For 55 to 57 percent of the validation patients outside the focal cell, one of the fifteen neighbours of the focal seed lies closer than their own nearest seed, and these patients contribute 43 to 48 percent of the random allocation’s improvement, about twice the contribution of the focal cell. Leakage thus arises mainly where the design expected none.

The reason is the scale of the seed cells. Five randomly drawn seeds leave validation patients at a mean distance of 0.39 from their nearest seed, so a cell is not a tight neighbourhood but a large share of the class. The fifteen patients nearest to one seed spread over a region that lies closer to many validation patients than their own seed does. The same diagnostic shows how weakly the coverage distance separates cohorts by composition: twenty patients consisting of the focal seed and its nineteen nearest neighbours cover validation patients at 0.346 to 0.356,

Cancer type	Coverage distance			κ_c	Focal-site share, %	
	r_D	r_C	r_R		Concentrated	Random
Adrenocortical carcinoma	0.309	0.258	0.246	0.81	72	71
Bladder urothelial carcinoma	0.573	0.479	0.430	0.66	12	6
Brain lower grade glioma	0.261	0.219	0.196	0.65	24	8
Breast invasive carcinoma	0.541	0.468	0.430	0.66	18	11
Cervical squamous and endocervical carcinoma	0.521	0.442	0.383	0.57	11	10
Colon adenocarcinoma	0.310	0.270	0.245	0.62	24	5
Esophageal carcinoma	0.560	0.466	0.410	0.62	22	7
Glioblastoma multiforme	0.324	0.270	0.245	0.69	45	13
Head and neck squamous cell carcinoma	0.359	0.301	0.274	0.69	21	12
Kidney chromophobe	0.203	0.159	0.153	0.86	18	15
Kidney renal clear cell carcinoma	0.256	0.218	0.191	0.59	36	18
Kidney renal papillary cell carcinoma	0.355	0.297	0.276	0.73	9	3
Liver hepatocellular carcinoma	0.368	0.309	0.285	0.71	32	19
Lung adenocarcinoma	0.500	0.429	0.385	0.62	21	7
Lung squamous cell carcinoma	0.499	0.424	0.394	0.71	16	5
Mesothelioma	0.512	0.421	0.398	0.79	8	5
Ovarian serous cystadenocarcinoma	0.441	0.346	0.341	0.95	56	44
Pancreatic adenocarcinoma	0.438	0.356	0.344	0.87	13	6
Pheochromocytoma and paraganglioma	0.249	0.183	0.161	0.76	68	69
Prostate adenocarcinoma	0.189	0.167	0.153	0.61	16	9
Rectum adenocarcinoma	0.272	0.221	0.199	0.70	20	13
Sarcoma	0.379	0.325	0.287	0.59	40	26
Skin cutaneous melanoma	0.493	0.420	0.396	0.76	20	11
Stomach adenocarcinoma	0.564	0.461	0.416	0.70	27	16
Testicular germ cell tumours	0.274	0.245	0.198	0.38	48	39
Thymoma	0.318	0.260	0.223	0.61	13	9
Thyroid carcinoma	0.341	0.282	0.237	0.56	21	17
Uterine carcinosarcoma	0.483	0.397	0.395	0.98	17	13
Uterine corpus endometrial carcinoma	0.439	0.375	0.339	0.64	16	3
Uveal melanoma	0.239	0.179	0.182	1.04	35	25
Mean over classes	0.386	0.322	0.294	0.70	27	17

Table 4: Class-level leakage stays above the permitted 0.5 for every cancer type except testicular germ cell tumours. Coverage distances and leakage are averaged over the three splits and five draws; κ_c is computed from the class’s own coverage distances. The focal-site shares give the percentage of added patients that come from the tissue source site of the focal seed, for the concentrated and the random addition rule.

Split	Focal cell, %	Concentrated, %		Random, %		Other cells closer, %
		Focal	Other	Focal	Other	
1	17	21	48	18	82	57
2	21	24	43	20	80	56
3	23	29	45	25	75	55

Table 5: Most of the concentrated allocation’s coverage improvement comes from validation patients outside the focal cell, which the design assumed it would not reach. The focal cell holds the validation patients whose nearest seed is the focal seed. The middle columns give each cell’s contribution to an allocation’s coverage improvement as a percentage of the random allocation’s total improvement; the concentrated columns sum to 100%. The last column gives the percentage of other-cell validation patients whose nearest training patient is closer in the concentrated than in the deep allocation.

better than the five random seeds at 0.385. Under the patient-mean embedding, adding almost any patients shortens the nearest-patient distance for many held-out patients.

5.4 Alternative allocation rules

Because each manipulation check costs no classifier fits, the diagnostic also evaluated alternative allocation rules against the same criterion before any redesign is fixed. Two coverage-maximizing rules select from the training pool without using validation patients: *farthest-point selection* repeatedly adds the pool patient farthest from the current set, and *facility-location selection* repeatedly adds the pool patient that most lowers the mean distance from all pool patients to their nearest selected patient. For an upward contrast, the random allocation is the low arm, and the criterion requires it to attain at most half of the high arm’s improvement. A *fully concentrated* variant replaces the random seeds by the focal seed and its four nearest neighbours, so that the deep allocation is itself concentrated and must be fitted anew. Table 6 lists the leakage of every variant.

Three observations constrain any redesign. First, farthest-point selection covers held-out patients worse than random selection, with leakage above one, because it adds outlying patients whom few validation patients resemble. Second, facility-location selection covers better than random selection, at 0.269 to 0.273 against 0.292 to 0.296, but random selection still attains 80 percent of its improvement, so an upward contrast is as weak as the downward ones. Third, combining the concentrated and the facility-location arm roughly halves the leakage of the neighbour design but still stops at 0.53 to 0.59. All seven variants build a twenty-patient allocation on top of a five-patient one, and within this nested structure the step from five to twenty patients dominates every coverage change that the selection rule can add.

6 Discussion and conclusion

Whether the residual breadth benefit of TCGA-UT is a patient-coverage benefit remains open after a second stopped manipulation check. Concentrating all additions around one seed lowers leakage from 0.84–0.88 to 0.67–0.73, but no classifier was trained, so neither the coverage gain nor the within-concentration residual constrains the accounts of Table 2.

The two checks together yield a result about the design space. The diagnostic refutes the assumption on which the concentrated design rests: seed cells do not localize coverage, and validation patients outside the focal cell supply about twice as much of the concentrated allocation’s improvement as those inside it. Across seven nested variants, including upward contrasts

Low-coverage arm	High-coverage arm	Leakage per split			Cells ≤ 0.5
Seeds + neighbours of all seeds	Seeds + random	0.84	0.85	0.88	0 of 90
Seeds + neighbours of focal seed	Seeds + random	0.69	0.67	0.73	6 of 90
Seeds + random	Seeds + farthest-point	1.39	1.35	1.37	0 of 90
Seeds + random	Seeds + facility-location	0.80	0.80	0.80	1 of 90
Seeds + neighbours of focal seed	Seeds + facility-location	0.55	0.53	0.59	36 of 90
Focal seed + 19 neighbours	Focal seed + 4 neighbours + random	0.64	0.61	0.66	13 of 90
Focal seed + 19 neighbours	Focal seed + 4 neighbours + facility-location	0.56	0.53	0.57	32 of 90

Table 6: No nested allocation rule reaches the permitted leakage of 0.5 in any split. Leakage is the share of the high arm’s coverage improvement over the five-patient allocation that the low arm already attains, computed as in Equation (3) from coverage distances averaged over classes and draws. The first two rows are the neighbour design of Queisler (2026c) and the design of this report. Cells count class–split combinations at or below 0.5.

with coverage-maximizing rules, the best leakage is 0.53. Under the patient-mean embedding, any nested design that must add little coverage while quadrupling patients runs against the fact that quadrupling patients shortens the nearest-patient distance for most held-out patients, whichever patients are added. A clean within-concentration residual, which requires a step that raises patient count at nearly constant coverage, is therefore out of reach for this coverage measure.

The measure is not insensitive to composition, however. Its dynamic range between cohorts of equal size is large once no five-patient allocation has to be nested inside both arms: in the diagnostic, twenty patients forming one tight neighbourhood cover validation patients at 0.346 to 0.356, and twenty patients selected by facility location starting from the same focal seed cover them at 0.268 to 0.271. This difference is almost as large as the coverage change from five to twenty random patients. The question whether dissimilar patients help can thus still be asked directly, at a fixed patient count, even though the question whether similar patients help beyond effective support cannot be asked with nested allocations.

6.1 Follow-up: coverage contrast at fixed patient count

A follow-up experiment drops the deep arm and contrasts two twenty-patient allocations with eight patches each, drawn per split, class, and draw around the focal seed of this report. The *clustered* allocation contains the focal seed and its nineteen nearest pool patients; the *dispersed* allocation contains the focal seed and nineteen patients added by facility-location selection on the training pool. Both allocations share patients, patches per patient, budget, and effective support, and both are anchored at the same randomly drawn patient, so the contrast isolates how widely the cohort spreads. The *dispersion gain* $\Delta_S = A_{\text{dispersed}} - A_{\text{clustered}}$ is read with the patient-macro balanced accuracy, bootstrap, and one-point threshold used throughout; the design requires 30 fits.

The manipulation check changes its reference. Instead of a share of a nested improvement, it requires the coverage contrast to carry at least half of the coverage change that accompanies

the breadth gain,

$$\rho = \frac{r_{\text{clustered}} - r_{\text{dispersed}}}{r_{\text{D}} - r_{\text{R}}} \geq 0.5 \quad (6)$$

in all three splits, where r_{D} and r_{R} are the deep and random coverage distances of Table 3. This check was evaluated on the same embeddings before fixing the design, and it passes in every split (Table 7).

Split	$r_{\text{clustered}}$	$r_{\text{dispersed}}$	$r_{\text{D}} - r_{\text{R}}$	ρ	Classes with $\rho_c \geq 0.5$
1	0.353	0.271	0.091	0.90	24 of 30
2	0.356	0.268	0.092	0.95	23 of 30
3	0.346	0.268	0.093	0.84	21 of 30

Table 7: The clustered and dispersed twenty-patient allocations differ in coverage by 84 to 95 percent of the coverage change from five to twenty random patients, so the manipulation check of Equation (6) passes in every split. Coverage distances are averaged over the 30 classes and five draws; ρ_c is the class-level ratio computed from the class’s own coverage distances.

The contrast is also large in the cohorts themselves. The mean pairwise distance among the twenty patients is 0.35 to 0.37 for the clustered, 0.67 for the dispersed, and 0.59 to 0.60 for the random allocation, so the dispersed allocation is slightly more spread out than random selection and the clustered allocation far less. Eight classes fall below $\rho_c = 0.5$ on average, among them uterine carcinosarcoma and uveal melanoma, in which both rules reach the same coverage; their class-level contrasts are reported but the gate applies at split level.

If the coverage account holds, Δ_S should be of the order of the residual breadth benefit, because the contrast reproduces most of the coverage change of the breadth step; an interval above one point would establish that dissimilar patients carry information that similar ones do not. An interval within $[-1, 1]$ would show that patient-mean coverage does not carry the residual, and would point to the distribution of a patient’s patches as the next candidate. The design gives up the within-concentration residual, so it cannot show directly that count helps beyond coverage. It also inherits the limitation of Section 7 in stronger form: the clustered allocation over-represents one appearance of each cancer type, and 24 to 27 percent of its added patients share the focal seed’s tissue source site against 16 to 18 percent of the dispersed additions. A secondary analysis by each test patient’s distance to the focal seed, which separates a coverage gain from a shift of the class boundary towards one appearance, therefore belongs in the protocol.

7 Limitations

The concentrated allocation does more than withhold coverage. Sixteen of its twenty patients resemble one seed, so it also over-represents one appearance of the cancer type, and a classifier may shift its class boundary towards that appearance. A positive coverage gain can therefore reflect this imbalance within the class as well as missing coverage. The focal-cell strata separate the two only partly: both readings predict a gain in the other cells.

Neighbours of one seed may share its tissue source site, because site signatures are visible in deep features of TCGA slides (Howard et al., 2021). The reported site shares quantify this overlap, and the site gain of 0.84 points (Queisler, 2026d) gives a reference scale for how much of a coverage gain sites alone could explain.

Coverage is defined on patient means in one representation, so a null coverage gain excludes coverage only as measured here. The within-concentration residual inherits the effective-support surface, whose slope is estimated on random allocations and transferred to a concentrated one.

All results are conditional on frozen Virchow2 features, multinomial logistic regression, the three overlapping splits, and the tested allocation sizes.

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